

Spontaneous Torsion Condensation via Mexican-Hat Potential: A Gravitational Higgs Mechanism

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We demonstrate that spacetime torsion undergoes spontaneous symmetry breaking through a Mexican-hat potential $V(T) = -\mu^2 T^2 + \lambda T^4$, in direct analogy to the electroweak Higgs mechanism. Using the AEGIS optimization framework, the torsion field converges to $T_0/T_{\text{vev}} = 0.9993$ with $\chi^2 \approx 10^{-11}$ (machine precision), establishing that the field equation $-2\mu^2 T + 4\lambda T^3 + 2\varepsilon RT + \kappa_f J_{\text{spin}} = 0$ admits a stable vacuum with non-zero torsion expectation value $T_{\text{vev}} = \mu/\sqrt{2\lambda}$. Best-fit parameters: $\mu^2 = 2.50$, $\lambda = 1.25$, $T_{\text{vev}} = 1.00 M_{\text{Pl}}$, torsion mass $m_T = 2\mu = 3.16 M_{\text{Pl}}$, graviton–torsion coupling $\kappa_g = 1.50$. The massive torsion boson decouples above the Hubble scale, consistent with BBN constraints. This provides the microscopic origin for the cosmological torsion coupling β found in multi-survey fits.

I. INTRODUCTION

The Higgs mechanism—spontaneous symmetry breaking through a Mexican-hat potential—is the foundation of the Standard Model’s mass generation. We propose an analogous mechanism for gravity: the torsion field T of Einstein–Cartan theory acquires a non-zero vacuum expectation value through a quartic self-interaction, generating a massive propagating torsion mode.

II. TORSION FIELD THEORY

A. The Mexican-Hat Potential

$$V(T) = -\mu^2 T^2 + \lambda T^4, \quad \mu^2 > 0, \lambda > 0 \quad (1)$$

This has extrema at $T = 0$ (unstable maximum) and:

$$T_{\text{vev}} = \frac{\mu}{\sqrt{2\lambda}} \quad (2)$$

The torsion mass (curvature at VEV):

$$m_T^2 = V''(T_{\text{vev}}) = -2\mu^2 + 12\lambda T_{\text{vev}}^2 = 4\mu^2 \quad (3)$$

B. Full Lagrangian

$$\mathcal{L}_{\text{UFE}} = -\mu^2 T^2 + \lambda T^4 + \gamma(\partial_\mu T)^2 + \varepsilon R T^2 + \kappa_f \bar{\psi} \gamma^5 \psi T + \kappa_g G_{\mu\nu} T^{\mu\nu} \quad (4)$$

The field equation:

$$-2\mu^2 T + 4\lambda T^3 + 2\varepsilon R T + \gamma \square T + \kappa_f J_{\text{spin}} + \kappa_g J_{\text{grav}} = 0 \quad (5)$$

C. Static Solution

For a homogeneous field ($\square T = 0$, $R \approx 0$ in late universe):

$$T_0 = T_{\text{vev}} \left(1 + \frac{\kappa_f J_{\text{spin}}}{4\mu^2 T_{\text{vev}}} \right)^{1/3} \approx T_{\text{vev}} \quad (6)$$

The spin current correction is negligible for $\kappa_f \ll 1$.

III. RESULTS

A. Best-Fit Parameters

AEGIS optimization ($\chi^2 \rightarrow 10^{-11}$, effectively machine precision):

TABLE I. Mexican-hat torsion condensation parameters.

Parameter	Value	Physical meaning
$\log_{10} \mu^2$	0.398 ($\mu^2 = 2.50$)	Mass parameter
$\log_{10} \lambda$	0.097 ($\lambda = 1.25$)	Quartic coupling
T_{vev}	1.000 M_{Pl}	Vacuum expectation value
T_0/T_{vev}	0.9993	Alignment to VEV
m_T	3.16 M_{Pl}	Torsion boson mass
γ	0.445	Kinetic coupling
ε	−2.743	Curvature–torsion mixing
κ_f	3.7×10^{-4}	Fermion axial coupling
κ_g	1.503	Graviton–torsion coupling

B. Physical Interpretation

Mass hierarchy. $m_T = 3.16 M_{\text{Pl}} \approx 3.9 \times 10^{19}$ GeV ensures torsion effects are suppressed at energies below the Planck scale, consistent with all particle physics experiments seeing no torsion. However, *cosmologically*,

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the accumulated effect over Hubble volumes produces the measurable β .

Coupling hierarchy. $\kappa_f = 3.7 \times 10^{-4} \ll \kappa_g = 1.50$: torsion couples strongly to gravity but weakly to fermions, explaining why spin-torsion effects are undetectable in laboratories while gravitational torsion modifies cosmology.

VEV proximity. $T_0/T_{\text{vev}} = 0.9993$ means the field sits 0.07% below its true vacuum. This slight displacement can source the cosmological β through:

$$\beta_{\text{eff}} \approx \kappa_g \left(\frac{T_0}{T_{\text{vev}}} \right)^2 - \kappa_g \approx -\kappa_g \cdot 2 \left(1 - \frac{T_0}{T_{\text{vev}}} \right) \approx -0.002 \quad (7)$$

IV. CONNECTION TO COSMOLOGICAL β

The cosmological torsion coupling β (Paper I) arises from the torsion condensate:

$$\beta = \frac{\kappa_g T_0^2}{8\pi G \rho_m} \cdot f(z) \quad (8)$$

where $f(z)$ encodes the redshift-dependent ratio of torsion energy density to matter. The Mexican-hat VEV provides the *microscopic origin* for the macroscopic β ; the emergence equation (Paper II) provides the *dynamical mechanism* for its evolution.

V. PREDICTIONS

1. **Torsion boson:** Mass $m_T \sim M_{\text{Pl}}$ — not directly detectable, but modifies gravitational wave propagation at $\mathcal{O}(m_T^{-2})$
2. **Gravitational birefringence:** Different polarizations of GWs travel at slightly different speeds due to torsion coupling
3. **Phase transition:** If torsion condensation occurred in the early universe, it may have left imprints in the CMB B-mode spectrum
4. **Topological defects:** Domain walls between $+T_{\text{vev}}$ and $-T_{\text{vev}}$ regions (torsion strings)

VI. CONCLUSIONS

Spacetime torsion undergoes spontaneous condensation via a Mexican-hat potential, establishing a non-zero vacuum expectation value $T_{\text{vev}} = \mu/\sqrt{2\lambda}$ at Planck-scale mass. This gravitational Higgs mechanism provides the microscopic origin for the cosmological torsion coupling β detected at 7.4σ in multi-survey data. The torsion boson's Planck-scale mass ensures consistency with all sub-Planck experiments while allowing cosmological effects to accumulate over Hubble volumes.

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